GEOTECHNICAL ENGINEERING

UNIT - IV : CONSOLIDATION OF SOILS

Course Code: 23A01503T

Syllabus: CONSOLIDATION: Types of compressibility - Primary consolidation and secondary consolidation - Stress history of clay; e-p and e-log p curves - Normally consolidated soil, over consolidated soil and under consolidated soil - Pre-consolidation pressure and its determination - Terzaghi's 1-D consolidation theory - Coefficient of consolidation: square root time and logarithm of time fitting methods.

1. Introduction

Under a load the soil decreases in volume - this is called compression. When a structure is built on a clay layer the clay takes years to settle, whereas in sand the settlement is complete as soon as the load is applied.

- Compressibility = the property by which the volume of the soil decreases under the action of compressive load.
- Consolidation = the gradual reduction in volume of a fully saturated soil of low permeability due to the drainage of pore water under a sustained static load. It is a time-dependent process.
- Volume decrease in a soil is due to: (i) compression of solid grains and water (very small, ignored), (ii) compression of air in voids (partially saturated soils), (iii) expulsion of water (and air) from the voids - the main reason.
- Consolidation is important because it governs the settlement of structures on clay (leaning tower of Pisa, Mexico City palace, Bangkok).

Types of settlement (compressibility)

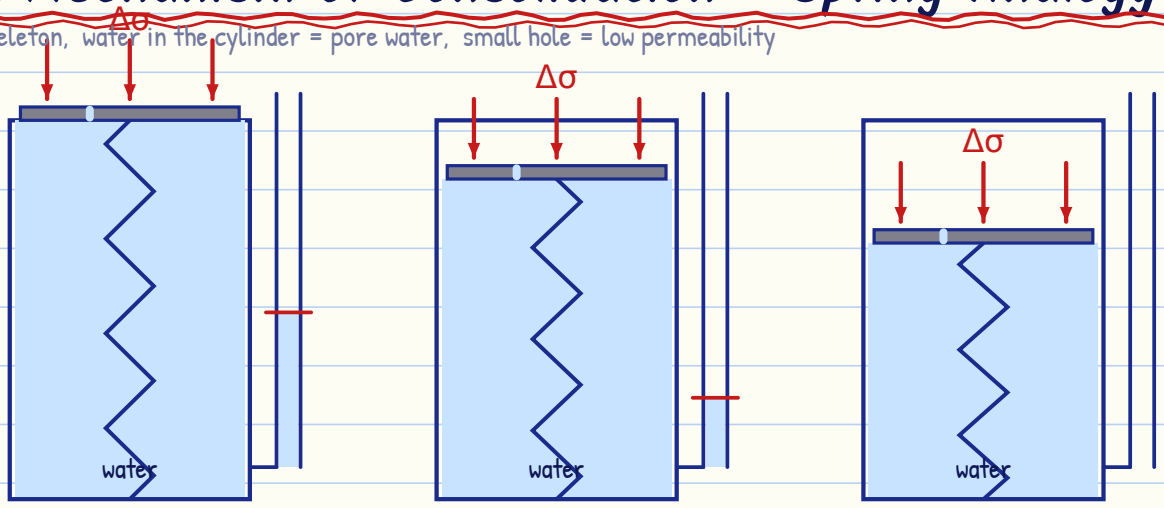
Type	Cause	Time
Immediate (initial) settlement S_i	elastic distortion of soil without change in water content (shape change)	immediately on loading
Primary consolidation S_c	expulsion of water from the voids, transfer of load from water to the soil skeleton	days to years
Secondary consolidation S_s	plastic readjustment of the particles at constant effective stress, after the excess pore pressure vanishes	continues for years

$$S_{total} = S_i + S_c + S_s$$

- For coarse-grained soils, consolidation is practically instant, so the settlement is mainly immediate. For saturated clays, S_c is the main component, and organic soil and peat have large S_s.

2. Mechanism of Consolidation - Spring Analogy

spring = soil skeleton, water in the cylinder = pore water, small hole = low permeability



At t = 0
u = Δσ, σ' = 0

0 < t < ∞
u + σ' = Δσ

t = ∞
u = 0, σ' = Δσ

Terzaghi explained the process by a piston-spring-cylinder model: a cylinder filled with water, a piston with a small hole (low permeability of clay) and a spring (soil skeleton).

- At t = 0: load Δσ is applied; water cannot escape instantly, is incompressible, so water takes the whole load: excess pore pressure u = Δσ, effective stress σ' = 0. The spring is not compressed.
- 0 < t < infinity: water escapes through the hole, the piston moves down, the spring compresses and carries a part of the load: σ' increases and u decreases, such that Δσ = σ' + u.
- At t = infinity: excess pore pressure fully dissipated, u = 0 and σ' = Δσ; the spring carries the whole load. Primary consolidation is complete.

Consolidation = dissipation of excess pore water pressure with a corresponding increase in effective stress. Rate of consolidation depends on the permeability and the drainage path length, not on the magnitude of the load.

3. Laboratory Consolidation Test (Oedometer)

IS 2720 (Part 15). Gives the e - log p curve, the compressibility parameters (C_c, a_v, m_v) and the coefficient of consolidation c_v.

- Apparatus: consolidometer (fixed ring or floating ring cell), loading frame (lever system), porous stones (top and bottom, for double drainage), dial gauge (0.002 mm least count), water bath.
- Specimen: undisturbed soil, in a ring of 60 mm dia (or 70 mm) and 20 mm height (ratio dia/height at least 3, to minimise side friction).
- Procedure: the specimen is kept in the cell with the stones, saturated by flooding. Load increments are applied with $\Delta p/p = 1$ (0.05, 0.1, 0.2, 0.4, 0.8, 1.6, 3.2 kg/cm^2), each load being kept for 24 hours (until the primary consolidation is complete). The dial readings are taken at 0.25, 1, 2.25, 4, 6.25, 9, 12.25, 16, 20.25, 25, 30.25 ... min (square numbers, for the root time method) and then at 24 h.
- After the maximum load the load is reduced in steps (unloading, rebound) and then reloaded to find the preconsolidation pressure (optional).

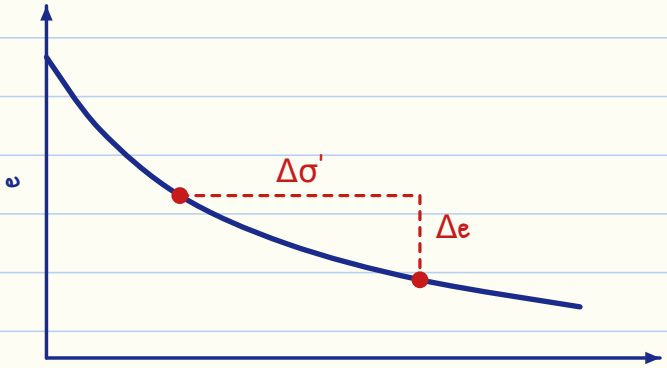
Calculation of void ratio

- Height of solids: $H_s = W_s / (A\ G\ \gamma_w)$. Initial void ratio $e_0 = (H_0 - H_s) / H_s$ (or $e_0 = w\ G$ for a saturated soil).

$$e = e_0 - \Delta H\ (1 + e_0) / H_0$$

- At the end of every load increment, find the final thickness H and the void ratio e: $e = (H - H_s) / H_s$. Plot e vs σ' (arithmetic or log scale).

4. e - p and e - log p Curves and Compressibility Parar



Coefficient of compressibility
$a_v = - \Delta e / \Delta \sigma'$ (m^2/kN)

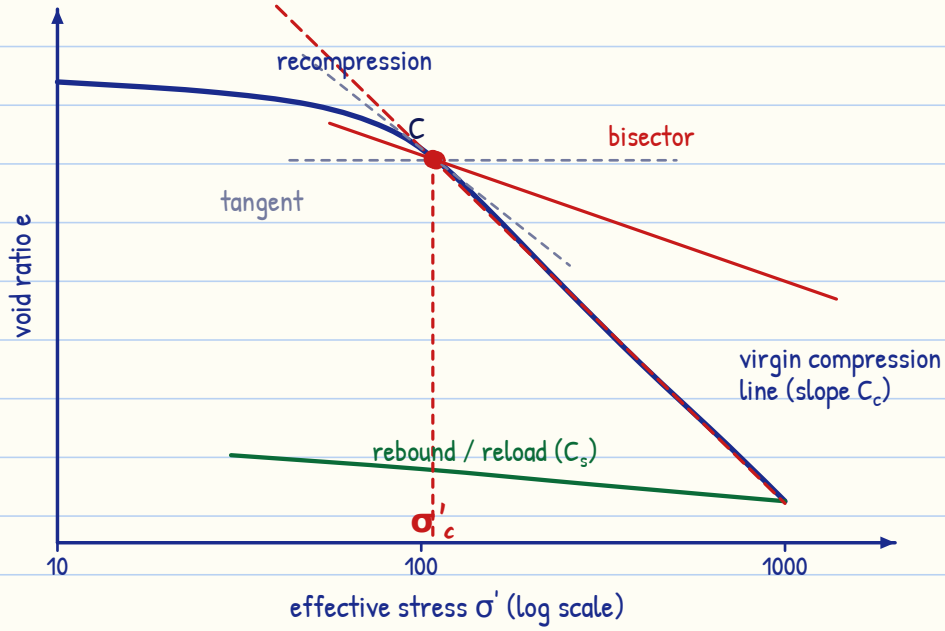
Coeff. of volume compressibility
$m_v = a_v / (1 + e_0)$

Compression index (e - log σ' plot)
$C_c = - \Delta e / \Delta \log \sigma'$

Swelling index C_s = 1/5 to 1/10 of C_c

- Coefficient of compressibility $a_v = - \Delta e / \Delta \sigma'$: slope of the e - p curve, decreases with increasing pressure; typical NC clay: 0.2 - 1 x 10^{-3} m^2/kN. (Note: negative sign because e decreases.)
- Coefficient of volume compressibility $m_v = a_v / (1 + e_0) = \Delta H / (H_0 \Delta \sigma') =$ volume change per unit initial volume per unit increase in pressure.

$$S_c = m_v \Delta \sigma' H_0 = [\, a_v / (1 + e_0) \,] \Delta \sigma' H_0 = [\, \Delta e / (1 + e_0) \,] H_0$$



- On the e - log σ' plot, the later part of the curve (beyond the preconsolidation pressure) is a straight line, the virgin compression line (VCL). Its slope is the compression index C_c.

$$C_c = - \Delta e / \Delta(\log_{10} \sigma') = (e_1 - e_2) / \log_{10} (\sigma'_2 / \sigma'_1)$$

- The unloading / reloading curve (recompression) has a flatter slope: the swelling (recompression) index C_s = 1/5 to 1/10 of C_c.

- Typical C_c: sand and gravel (very small), NC clays 0.2 – 0.5, soft clays 0.5 – 1.0, organic clay and peat > 1.0. High C_c means high compressibility.
- Empirical correlations: Terzaghi and Peck (undisturbed clay): $C_c = 0.009\,(w_L - 10)$; remoulded clay: $C_c = 0.007\,(w_L - 10)$. Also $C_c = 0.54\,(e_0 - 0.35)$ (Nishida) and $C_c = 0.0054\,(2.6\,w_n \ldots)$ approx.

Settlement of a normally consolidated clay layer

$$S_c = [\, C_c H / (1 + e_0) \,] \log_{10} [\, (\sigma'_0 + \Delta\sigma) / \sigma'_0 \,]$$

- H = thickness of the clay layer, σ'_0 = effective overburden pressure at the middle of the layer, $\Delta\sigma$ = increase in vertical stress at the middle of the layer.

Example 1

NC clay layer 4 m thick, $e_0 = 1.10$, $C_c = 0.30$, σ'_0 at the mid-depth = 80 kPa, $\Delta\sigma = 60$ kPa.

- $S_c = (0.30 \times 4 / 2.10) \times \log (140/80) = 0.571 \times 0.243 = 0.139$ m = 139 mm.

Example 2 (compression index)

A clay with $w_L = 52$ % (undisturbed): $C_c = 0.009\,(52 - 10) = 0.378$.

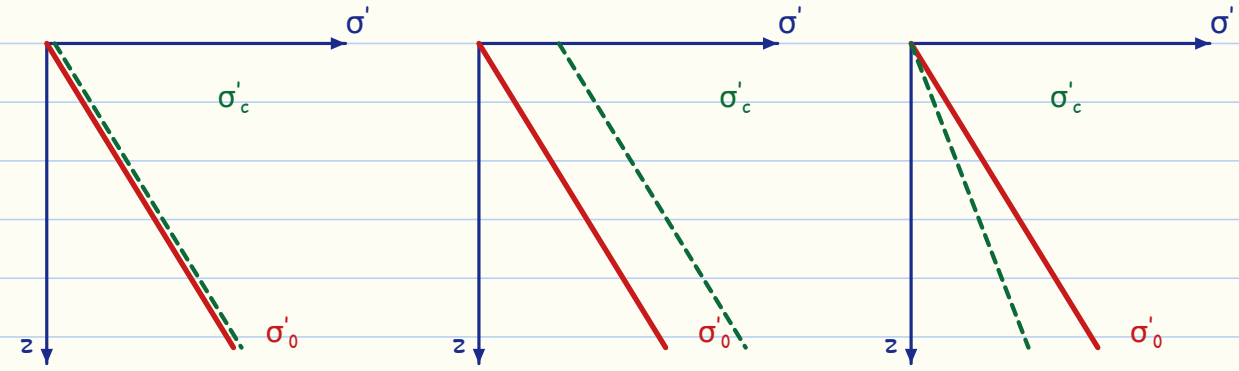
Example 3 (from void ratios)

Void ratio decreases from 0.92 to 0.80 when the pressure changes from 100 to 200 kPa. $a_v = 0.12/100 = 1.2 \times 10^{-3}$ m^2/kN; $C_c = 0.12 / \log 2 = 0.12/0.301 = 0.399$; $m_v = 1.2 \times 10^{-3} / 1.92 = 6.25 \times 10^{-4}$ m^2/kN.

5. Stress History of Clay

The stress history is the record of the maximum effective stress the soil has been subjected to in the past. The soil remembers the maximum past pressure.

- Preconsolidation pressure (σ'_c or p_c): the maximum effective stress experienced by the soil in its past.
- Over-consolidation ratio OCR = σ'_c / σ'_0, where σ'_0 = present effective overburden pressure.



Normally consolidated OCR = 1 | *Over-consolidated* OCR > 1 | *Under-consolidated* OCR < 1

(a) Normally consolidated (NC) soil

- Present effective overburden = the maximum past pressure: $\sigma'_0 = \sigma'_c$, OCR = 1. The soil has never been subjected to a pressure greater than the present one and is fully consolidated under the present overburden. Example: recently deposited alluvium and marine clay.

(b) Over-consolidated (OC) soil

- $\sigma'_0 < \sigma'_c$: OCR > 1. The soil was subjected in the past to a greater pressure than the present.
- Causes: erosion of overburden, melting of glaciers (ice load removed), lowering of the water table, desiccation (drying and shrinkage - crust formation), tectonic pressures, excavation, and secondary compression (ageing).
- Behaviour: stiffer, less compressible, higher strength, lower void ratio than an NC clay at the same depth. On loading the soil follows the flat recompression curve (C_s) until σ'_c and then the VCL.

(c) Under-consolidated (UC) soil

- The soil is still consolidating under the present overburden; excess pore pressure not yet dissipated, so $\sigma'_0 > \sigma'_c$ (OCR < 1). Example: recently placed fills and rapidly deposited sediments, e.g. delta deposits, dredged fill; highly compressible and weak.

State	OCR	σ'_0 compared with σ'_c	Remark
Normally consolidated	= 1	equal	on the VCL
Over-consolidated	> 1	$\sigma'_0 < \sigma'_c$	stiff, small settlement
Under-consolidated	< 1	$\sigma'_0 > \sigma'_c$	settling under own weight

Settlement of an over-consolidated clay

- If $\sigma'_0 + \Delta\sigma <= \sigma'_c$ (the final pressure remains below the preconsolidation pressure):

$$S_c = [\, C_s H / (1 + e_0) \,] \log [\, (\sigma'_0 + \Delta\sigma) / \sigma'_0 \,]$$

- If $\sigma'_0 < \sigma'_c < \sigma'_0 + \Delta\sigma$ (crosses the preconsolidation pressure):

$$S_c = [C_s H/(1+e_0)] \log (\sigma'_c/\sigma'_0) + [C_c H/(1+e_0)] \log [(\sigma'_0+\Delta\sigma)/\sigma'_c]$$

Example 4

OC clay 3 m thick, $e_0 = 0.95$, $C_c = 0.28$, $C_s = 0.05$, $\sigma'_0 = 60$ kPa, $\sigma'_c = 100$ kPa, $\Delta\sigma = 80$ kPa. Final pressure 140 kPa > σ'_c, so:

- $S_c = (0.05 \times 3/1.95) \log (100/60) + (0.28 \times 3/1.95) \log (140/100) = 0.0769 \times 0.2218 + 0.4308 \times 0.1461 = 0.01706 + 0.06294 = 0.080$ m = 80 mm. OCR = 100/60 = 1.67.

6. Preconsolidation Pressure and its Determination

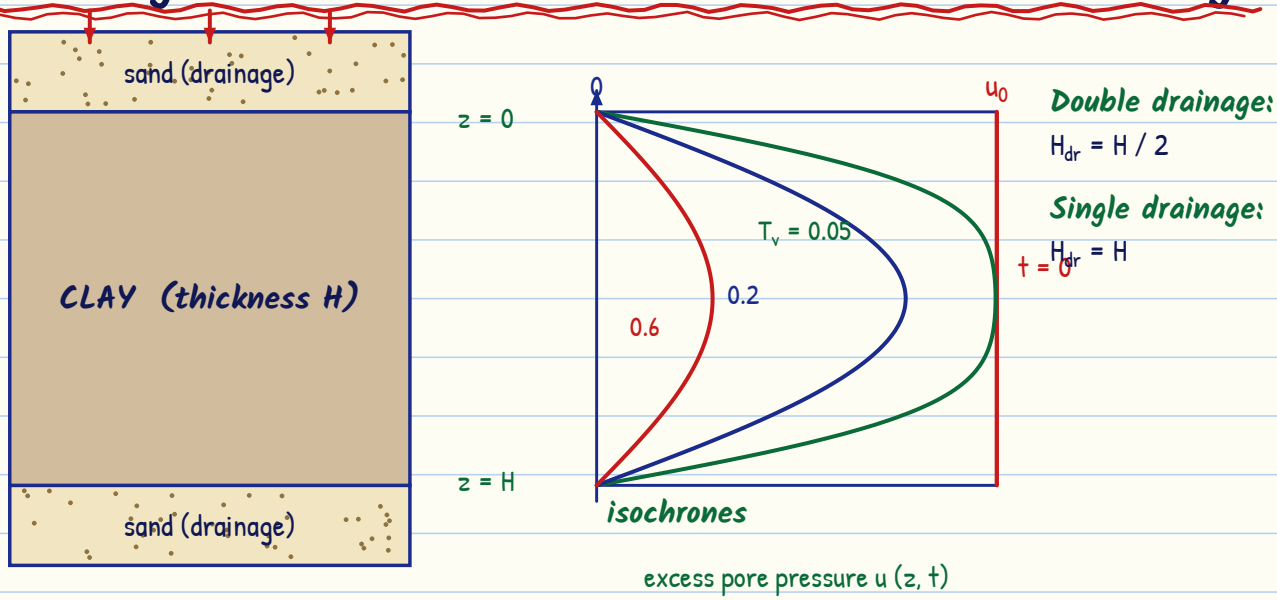
Casagrande's graphical method (1936) on the e - log σ' curve (see the figure above):

- Step 1: Pick the point C of maximum curvature (smallest radius) on the e - log p curve.
- Step 2: Draw the horizontal line through C and the tangent at C.
- Step 3: Draw the bisector of the angle between the horizontal line and the tangent.
- Step 4: Extend the straight line portion of the virgin compression curve backwards till it meets the bisector.
- Step 5: The abscissa of the point of intersection gives the preconsolidation pressure σ'_c.
- Other methods: Schmertmann's construction, Burmister, and the work-per-unit-volume (Becker) method.

Notes

- Sample disturbance rounds off the curve and so reduces the apparent σ'_c and flattens the VCL - the Schmertmann method corrects for it.
- OCR is found as σ'_c/σ'_0; the soil is NC if σ'_c is about equal to σ'_0.
- In design, if the structure pressure exceeds σ'_c the settlement will be large (use the VCL slope C_c).

7. Terzaghi's One-Dimensional Consolidation Theory



Terzaghi (1925) gave the theory for the rate of consolidation of a saturated clay layer under a sudden load, with drainage and compression in the vertical direction only.

Assumptions

- The soil is homogeneous and completely saturated.
- Soil grains and the water are incompressible.
- Compression and flow are one-dimensional (vertical); lateral strain is zero.
- Darcy's law is valid; k is constant during consolidation.
- m_v (and hence c_v) is constant; the relation between e and σ' is linear (for small stress increments).
- The strains are small; the load is applied instantaneously and stays constant; time lag is due entirely to the low permeability of soil (no secondary compression).

Differential equation

For an element of thickness dz, the rate of volume decrease = net outflow of water: $\partial(e)/\partial t$ ... leads to

$$\partial u / \partial t = c_v \, \partial^2 u / \partial z^2 \qquad c_v = k / (m_v \gamma_w) = k (1 + e_0) / (a_v \gamma_w)$$

- c_v = coefficient of consolidation (cm^2/s or $m^2/year$): indicates how fast the pore pressure dissipates. Large for sand and silt, small for clay (10^{-3} to 10^{-4} cm^2/s).
- Solution by Fourier series with $u = u_0$ at $t = 0$, and $u = 0$ at the drainage boundaries:

$$u = \Sigma \, (2 u_0 / M) \sin (M z / H_{dr}) \, e^{-M^2 T_v} \qquad M = (\pi/2)(2m + 1), \; m = 0, 1, 2, \ldots$$

- Time factor $T_v = c_v t / H_{dr}^2$ (dimensionless). H_{dr} = longest drainage path = H/2 for double drainage (sand above and below) and H for single drainage (one drainage face).

Degree of consolidation

- $U_z = (u_0 - u_z) / u_0 = 1 - u_z / u_0$ at a depth z. Average degree of consolidation U (whole layer):

$$U = S_t / S_c = 1 - \Sigma\, (2/M^2)\, e^{-M\hat{}\{2\}\, T_v\}}$$

- U depends only on T_v (a single curve for a given drainage and initial pressure distribution).
- Approximations: for $U < 60$ %, $T_v = (\pi/4)\, U^2$; for $U > 60$ %, $T_v = 1.781 - 0.933 \log_{10} (100 - U\%)$.

U (%)	10	20	30	40	50	60	70	80	90	100
T_v	0.008	0.031	0.071	0.126	0.197	0.287	0.403	0.567	0.848	infinity

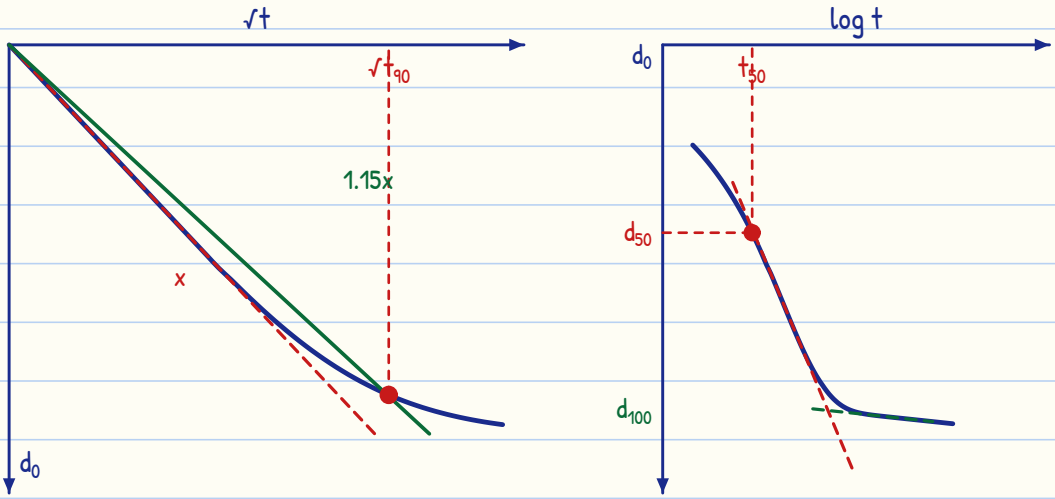
- Settlement at time t: $S_t = U \times S_c$. Time: $t = T_v\, H_{dr}^2 / c_v$. Time for the consolidation varies with the square of the drainage path; a layer with double drainage consolidates 4 times faster than with single drainage.

Example 5

Clay layer 4 m thick, double drainage, $c_v = 4 \times 10^{-3}$ cm^2/s, $S_c = 139$ mm.

- $H_{dr} = 2$ m $= 200$ cm. For $U = 90$ %, $T_v = 0.848$: $t_{90} = 0.848 \times 200^2 / (4 \times 10^{-3}) = 8.48 \times 10^6$ s = 98 days.
- For $U = 50$ %, $T_v = 0.197$: $t = 0.197 \times 40000 / 0.004 = 1.97 \times 10^6$ s = 22.8 days; $S_t = 0.5 \times 139 =$ 69.5 mm.
- If single drainage, $H_{dr} = 4$ m and $t_{90} = 4 \times 98 =$ 392 days.

8. Coefficient of Consolidation c_v - Curve Fitting Metho



Square root of time method *Log time method*

The c_v is found from the laboratory time - compression readings (dial reading vs time) of a load increment by two fitting methods, which match the theoretical U - T_v curve with the experimental curve.

(a) Square root of time method (Taylor, 1942)

- Plot the dial reading (compression) versus the square root of time ($min^{0.5}$). The initial part is a straight line ($U < 60$ %, $T_v = (\pi/4)U^2$, so the compression is proportional to sqrt(t)).
- Extend the initial straight line back to $t = 0$ to find the corrected zero d_0.
- Draw a second line from d_0 with all abscissae 1.15 times those of the first line. It cuts the experimental curve at the point corresponding to 90 % consolidation, which gives sqrt(t_{90}).

$$c_v = 0.848\ H_{dr}^2 / t_{90}$$

- (Since the theoretical ratio of abscissa at 90 % to the line is 1.15.) $d_{100} = d_0 + (10/9)(d_{90} - d_0)$.

(b) Logarithm of time method (Casagrande)

- Plot the dial reading versus log t. The curve is S-shaped (reverse) with a straight line in the middle part.
- Corrected zero d_0: choose two times t_1 and $t_2 = 4\ t_1$ (in the initial parabolic part); the vertical difference x between the readings gives the zero: $d_0 = d_1 - x$ (the reading at t_1 minus x).

- d_{100} = ordinate at the intersection of the tangent to the middle straight portion and the tangent to the final (secondary) portion.
- $d_{50} = (d_0 + d_{100})/2$; the time at d_{50} is t_{50}.

$$c_v = 0.197\ H_{dr}^2 / t_{50}$$

Calculations with the results

- H_{dr} is the average drainage path of the specimen for the load increment = half the average thickness (double drainage): $H_{dr} = (H_i + H_f)/4$.
- Then $k = c_v\ m_v\ \gamma_w$ - gives the coefficient of permeability of the clay.

Example 6 (root time)

Specimen 20 mm thick, double drained, H_{dr} = 1.0 cm. $\sqrt{t_{90}}$ = 3.4 $\sqrt{}$min, so t_{90} = 11.56 min = 694 s.

$c_v = 0.848 \times (1.0)^2 / 694 = 1.22 \times 10^{-3}$ cm^2/s.

Example 7 (log time)

t_{50} = 3.0 min = 180 s, H_{dr} = 1.0 cm: $c_v = 0.197 \times 1 / 180 = 1.09 \times 10^{-3}$ cm^2/s.

Comparison

Square root of time (Taylor)	Log of time (Casagrande)
uses t_{90}, T_v = 0.848	uses t_{50}, T_v = 0.197
gives a good initial straight line	needs the secondary compression to be clear
better for soils with little secondary compression	better for soils showing a distinct secondary portion
$c_v = 0.848\ H^2/t_{90}$	$c_v = 0.197\ H^2/t_{50}$

9. Secondary Consolidation

After the primary consolidation (u = 0), the soil continues to compress slowly at constant effective stress due to plastic adjustment (creep) of the soil grains and the adsorbed water layers.

- On the log t plot it appears as a straight line of small slope after d_{100}.

$$C_\alpha = \Delta e / \Delta \log_{10} t \qquad S_s = [\ C_\alpha\ H / (1 + e_p)\ ] \log_{10} (t_2 / t_1)$$

- e_p = void ratio at the end of the primary consolidation; t_1 = time at the end of primary; t_2 = time at which S_s is required.
- C_α / C_c = 0.03 - 0.06 for most clays, up to 0.1 for organic soils and peat. S_s is large for organic soils, peat, and highly plastic soft clays; small for over-consolidated clays.

- Secondary compression is not related to the dissipation of pore pressure (Terzaghi's theory neglects it).

10. Quick Revision

Item	Formula
Void ratio	$e = e_0 - \Delta H (1 + e_0) / H_0$
Coefficient of compressibility	$a_v = - \Delta e / \Delta \sigma'$
Volume compressibility	$m_v = a_v / (1 + e_0)$
Compression index	$C_c = - \Delta e / \Delta \log \sigma' = 0.009 (w_L - 10)$
Settlement (NC)	$S_c = [C_c H / (1 + e_0)] \log[(\sigma'_0 + \Delta\sigma)/\sigma'_0]$
OCR	$OCR = \sigma'_c / \sigma'_0$
Coefficient of consolidation	$c_v = k / (m_v \gamma_w)$
Time factor	$T_v = c_v t / H_{dr}^2$
Square root time	$c_v = 0.848 H_{dr}^2 / t_{90}$
Log time	$c_v = 0.197 H_{dr}^2 / t_{50}$
Secondary settlement	$S_s = [C_\alpha H / (1 + e_p)] \log (t_2/t_1)$

Important questions (exam)

- Distinguish between compaction and consolidation. Explain the spring analogy for primary consolidation.
- Explain immediate, primary and secondary settlements.
- Explain e - p and e - log p curves. Define a_v, m_v, C_c, C_s.
- What is stress history? Explain NC, OC and UC clays. How do you find the preconsolidation pressure?
- State the assumptions of Terzaghi's theory and derive the equation $\partial u/\partial t = c_v \partial^2 u/\partial z^2$.
- Describe the square root of time and the log of time fitting methods for c_v.